

Human Face Recognition Mini Project - Full Guide

Introduction

Human Face Recognition is a computer vision project used to detect human faces from images. This project uses Python and OpenCV library. It is easy to understand and suitable for college practical exams.

Step 1: Install Python

Download Python from the official Python website.

Important:

While installing Python tick the option:
Add Python to PATH

Step 2: Install OpenCV

Open Command Prompt and write:

```
py -m pip install opencv-python
```

Step 3: Create Project Folder

Create a folder named:
FaceRecognitionProject

Inside the folder save:

- project.py
- person.jpg

Step 4: Save Image

Save any face image inside project folder with the name:
person.jpg

Step 5: Python Program

```
import cv2

# Load image
img = cv2.imread("person.jpg")

# Load face detector
face_cascade = cv2.CascadeClassifier(
    cv2.data.haarcascades + 'haarcascade_frontalface_default.xml'
)

# Convert image into gray
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

```

# Detect faces
faces = face_cascade.detectMultiScale(gray, 1.3, 5)

# Draw rectangle around face
for (x, y, w, h) in faces:
    cv2.rectangle(img,
                  (x, y),
                  (x+w, y+h),
                  (0, 255, 0),
                  2)

# Show output image
cv2.imshow("Face Recognition", img)

# Wait until key press
cv2.waitKey(0)

# Close window
cv2.destroyAllWindows()

```

Step 6: Run Program

Open Command Prompt inside project folder.

Write:

python project.py

or

py project.py

Output

The image will open and the detected face will have a green rectangle around it.

Code Explanation

1. import cv2
Used to import OpenCV library.
2. cv2.imread()
Used to load image.
3. Haar Cascade Classifier
Used for detecting faces.
4. cv2.cvtColor()
Converts image into grayscale.
5. detectMultiScale()
Detects faces from image.
6. cv2.rectangle()
Draws rectangle around face.
7. cv2.imshow()
Displays final output image.

Folder Structure

FaceRecognitionProject



■■■■ project.py

■■■■ person.jpg

Advantages

- Easy to implement
- Fast face detection
- Useful for beginners
- Simple practical project

Disadvantages

- Less accurate in low light
- Cannot identify person name

Applications

- Attendance System
- Mobile Face Unlock
- Security Systems
- CCTV Surveillance

Viva Questions and Answers

Q1. What is OpenCV?

OpenCV is an image processing library.

Q2. What is Face Recognition?

Detecting human face from image or video.

Q3. Which algorithm is used?

Haar Cascade Algorithm.

Q4. Why grayscale image is used?

To reduce complexity and increase speed.

Q5. What is cv2?

It is OpenCV module in Python.

Conclusion

This mini project successfully demonstrates Human Face Recognition using Python and OpenCV. It helps students understand basic image processing and face detection concepts.